

1 Formule de cuadratura

Forma generală (unde w este o funcție pozitivă numită pondere):

$$\int_a^b w(x) f(x) dx = \sum_{i=0}^n A_i f(x_i) + R(f) \quad (1)$$

unde A_i, x_i nu depind de f . Folosirea:

$$\int_a^b w(x) f(x) dx \approx \sum_{i=0}^n A_i f(x_i)$$

$x_i \in [a, b]$, distincte, se numesc nodurile f.c., A_i ponderile (coef.) f.c. iar $R(f)$ este restul f.c..

Definition 1 Formula de cuadratură (1) are grad de exactitate m dacă:

$$\begin{aligned} R(e_k) &= 0, k = \overline{0, m} \\ R(e_{m+1}) &\neq 0 \end{aligned}$$

unde:

$$e_k : \mathbb{R} \rightarrow \mathbb{R}, e_k(x) = x^k$$

F.c. de tip interpolator:

$$\begin{aligned} \int_a^b w(x) f(x) dx &= \int_a^b w(x) L(x_0, \dots, x_n; f)(x) dx + R(f) = \\ &= \int_a^b w(x) \sum_{i=0}^n l_i(x) f(x_i) + R(f) = \\ &= \sum_{i=0}^n \left(\int_a^b w(x) l_i(x) dx \right) f(x_i) + R(f) \\ A_i &= \int_a^b w(x) l_i(x) dx \end{aligned}$$

Theorem 2 Formula de cuadratură este de tip interpolator dacă și numai dacă gradul de exactitate este mai mare sau egal cu n .

Pt. noduri echidistante: $x_i = a + \frac{b-a}{n}i$. Cazuri particulare:

1. $n = 1$ $w(x) = 1$ orice $x \in [a, b]$:

$$\begin{aligned} \int_a^b f(x) dx &= A_0 f(a) + A_1 f(b) + R(f) \\ \int_a^b L(a, b; f)(x) dx &= A_0 f(a) + A_1 f(b) = \\ &= \frac{(b-a)}{2} (f(a) + f(b)) \end{aligned}$$

formula trapezului:

$$\int_a^b f(x) dx = \frac{(b-a)}{2} (f(a) + f(b)) + R(f)$$

$$R(f) = R(e_2) \frac{f''(\xi)}{2!}$$

$$R(e_2) = \int_a^b x^2 dx - \frac{b-a}{2} (a^2 + b^2) = \frac{1}{6} (a-b)^3$$

2. Formula lui Simpson ($n = 2$):

$$\int_a^b f(x) dx = A_0 f(a) + A_1 f\left(\frac{a+b}{2}\right) + A_2 f(b) + R(f) =$$

$$= \frac{b-a}{6} \left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right) + R(f)$$

$$R(f) = R(e_4) \frac{f^{iv}(\xi)}{4!} = \frac{1}{120} (a-b)^5 \frac{f^{iv}(\xi)}{4!} (e_4(x) := x^4)$$

$$120 * 4! = : 2880$$

$$R(e_4) = \int_a^b x^4 dx - (b-a)/6 \left(a^4 + 4 \left(\frac{a+b}{2} \right)^4 + b^4 \right) = \frac{1}{120} (a-b)^5$$

3. Formula dreptunghiului ($n = 0$?) $x_0 = \frac{a+b}{2}$

$$\int_a^b f(x) dx = \frac{b-a}{1} f\left(\frac{a+b}{2}\right) + R(f)$$

$$R(e_1) = \int_a^b x dx - (b-a) \frac{a+b}{2} = 0$$

$$R(e_2) = \int_a^b x^2 dx - (b-a) \left(\frac{a+b}{2} \right)^2 = \frac{(b-a)^3}{12}$$

$$R(f) = \frac{(b-a)^3}{12} \frac{f''(\xi)}{2!}$$

4. Formule compuse:

(a) Formula trapezelor (se obține integrând funcția spline de interpolare de grad 1):

$$\int_a^b f(x) dx = \frac{b-a}{2n} \left(f(a) + f(b) + 2 \sum_{k=1}^{n-1} f\left(a + \frac{b-a}{n} k\right) \right) + R(f)$$

$$\text{unde } R(f) = \frac{K}{n^2}.$$

(b) Formula compusă a lui Simpson:

$$\int_a^b f(x) dx = \frac{b-a}{6n} \left(f(a) + f(b) + 2 \sum_{k=1}^{n-1} f\left(a + \frac{b-a}{n}k\right) + 4 \sum_{k=1}^{n-1} f\left(a + \frac{b-a}{2n}(2k-1)\right) \right) + R$$

formula lui Simpson pe intervalul $\left[a + \frac{b-a}{n}(k-1), a + \frac{b-a}{n}k\right]$

$$\int_{a+\frac{b-a}{n}(k-1)}^{a+\frac{b-a}{n}k} f(x) dx = \frac{b-a}{6n} \left(f\left(a + \frac{b-a}{n}(k-1)\right) + f\left(a + \frac{b-a}{n}k\right) + 4f\left(a + \frac{b-a}{2n}(2k-1)\right) \right)$$

(c) formula dreptunghiurilor:

$$\int_a^b f(x) dx = \frac{b-a}{n} \sum_{k=1}^n f\left(a + \frac{b-a}{2n}(2k-1)\right) + R(f)$$

2 Formule de cuadratură de tip Gauss

Aleg A_i și x_i , $i = \overline{0, n}$ astfel încât f.c. să aibă grad de exact. maxim $(2n+1)$:

$$\begin{aligned} R(e_k) &= \int_a^b w(x) x^k dx - \sum_{i=0}^n A_i x_i^k = 0 \\ k &= \overline{0, 2n+1} \end{aligned}$$

Exemplu: Să se det. o f.c. cu grad maxim de exact. pe 2 noduri, intervalul $[-1, 1]$, $w(x) = \sqrt{1-x}$.

Deci $n = 1$,

$$\int_{-1}^1 \sqrt{1-x} f(x) dx = A_0 f(x_0) + A_1 f(x_1) + R(f)$$

sistemul:

$$\begin{cases} \int_{-1}^1 \sqrt{1-x} 1 dx = A_0 1 + A_1 1 = \frac{4}{3} \sqrt{2} \\ \int_{-1}^1 \sqrt{1-x} x dx = A_0 x_0 + A_1 x_1 = -\frac{4}{15} \sqrt{2} \\ \int_{-1}^1 \sqrt{1-x} x^2 dx = A_0 x_0^2 + A_1 x_1^2 = \frac{44}{105} \sqrt{2} \\ \int_{-1}^1 \sqrt{1-x} x^3 dx = A_0 x_0^3 + A_1 x_1^3 = -\frac{52}{315} \sqrt{2} \end{cases}$$

$$\begin{cases} A_0 1 + A_1 1 = \frac{4}{3} \sqrt{2} \\ A_0 x_0 + A_1 x_1 = -\frac{4}{15} \sqrt{2} \\ A_0 x_0^2 + A_1 x_1^2 = \frac{44}{105} \sqrt{2} \\ A_0 x_0^3 + A_1 x_1^3 = -\frac{52}{315} \sqrt{2} \end{cases}, \text{ Solution is: } [A_0 = \frac{21}{100} \sqrt{2} \left(\frac{4}{63} \sqrt{7} \sqrt{10} - \frac{1}{9} \right) + \frac{69}{100} \sqrt{2}, A_1 = \frac{193}{300} \sqrt{2} - \frac{21}{100} \sqrt{2}$$

$$[A_0 = \frac{69}{100} \sqrt{2} - \frac{21}{100} \sqrt{2} \left(\frac{4}{63} \sqrt{7} \sqrt{10} + \frac{1}{9} \right), A_1 = \frac{21}{100} \sqrt{2} \left(\frac{4}{63} \sqrt{7} \sqrt{10} + \frac{1}{9} \right) + \frac{193}{300} \sqrt{2}, x_0 = \frac{4}{63} \sqrt{7} \sqrt{10} - \frac{1}{9}, x_1 = -$$

$$, \text{ Solution is: } \begin{cases} [A_0 = 0.78505, A_1 = 1.1006, x_0 = 0.4201, x_1 = -0.64232], \\ [A_0 = 1.1006, A_1 = 0.78505, x_0 = -0.64232, x_1 = 0.4201] \end{cases}$$

Formula de c. este:

$$\int_{-1}^1 \sqrt{1-x} f(x) dx = 0.78505 f(0.4201) + 1.1006 f(-0.6432) + R(f)$$

$$R(f) = R(e_4) \frac{f^{iv}(\xi)}{4!}$$

$$e_k : \mathbb{R} \rightarrow \mathbb{R} \quad e_k(x) = x^k$$

$$f(x) = \sqrt{1+x}$$

$$\int_{-1}^1 \sqrt{1-x^2} dx = \frac{1}{2} \pi = 1.5708$$

$$0.78505 \sqrt{0.4201+1} + 1.1006 \sqrt{-0.6432+1} = 1.5929$$

Theorem 3 Dacă $f.c.$ e de tip Gauss atunci nodurile sunt rădăcinile pol. ortogonal P_{n+1} (de grad $n+1$) pe intervalul $[a, b]$ cu ponderea w .

ortogonal P_{n+1} (de grad $n+1$) pe intervalul $[a, b]$ cu ponderea w :

$$\int_a^b w(x) P_{n+1}(x) x^k dx = 0, k \leq n.$$

Remark 4

$$A_i = ? = \int_a^b w(x) l_i^2(x) dx = \int_a^b w(x) l_i(x) dx$$

$$l_i(x) = \frac{P_{n+1}(x)}{(x-x_i) P'_{n+1}(x_i)}$$

Example 5 $\int_{-1}^1 \sqrt{1-x} f(x) dx = A_0 f(x_0) + A_1 f(x_1) + R(f)$; $P_2 = ?$

Ideea : Gramm-Schmidt: Să se determine polinomul ortogonal de grad n pe $[a, b]$ cu ponderea w . De fapt se determină toate polinoamele ortogonale până la gradul n .

$$P_0(x) = 1;$$

pentru $k = 1$ până la n se det, polinomul P_k :

$$P_k(x) = x^k + \sum_{i=0}^{k-1} \alpha_{k,i} P_i(x) \mid * w(x) P_i(x), \int_a^b$$

$$\alpha_{k,i} = - \frac{\int_a^b w(x) P_i(x) x^k dx}{\int_a^b w(x) P_i^2(x) dx}, \text{ pentru } i = \overline{0, k-1}.$$

$$\begin{aligned}
P_0(x) &= 1 \\
P_1(x) &= x + \alpha_{1,0}P_0(x) : \alpha_{1,0} = 1/5 \text{ rezultă din } \int_{-1}^1 \sqrt{1-x} \\
P_2(x) &= x^2 + \alpha_{2,1}P_1(x) + \alpha_{2,0}P_0(x) \\
\int_{-1}^1 \sqrt{1-x} (x^2 + \alpha_{2,1}P_1(x) + \alpha_{2,0}P_0(x)) P_0(x) dx &= 0
\end{aligned}$$

$$\int_{-1}^1 \sqrt{1-x} (x^2 + \alpha_{2,1}(x + 1/5) + \alpha_{2,0}1) 1 dx = 0$$

$$\alpha_{2,0} = -\frac{\int_{-1}^1 \sqrt{1-x} x^2 1 dx}{\int_{-1}^1 \sqrt{1-x} 1^2 dx}$$

$$\int_{-1}^1 \sqrt{1-x} (x^2 + \alpha(x + 1/5) + \beta 1) (x + 1/5) dx = 0$$

$$\alpha_{2,1} = -\frac{\int_{-1}^1 \sqrt{1-x} x^2 (x + 1/5) dx}{\int_{-1}^1 \sqrt{1-x} (x + 1/5)^2 dx}$$

$$\begin{aligned}
&\int_{-1}^1 \sqrt{1-x} x^2 (x + 1/5) dx \stackrel{\substack{\sqrt{1-x}=t \\ x=1-t^2}}{=} \int_{\sqrt{2}/2}^0 t (1-t^2)^2 (1-t^2 + 1/5) (-2t) dt = \\
&- \frac{128}{1575} \sqrt{2} ; \int_{-1}^1 \sqrt{1-x} (x + 1/5) dx \stackrel{\sqrt{1-x}=t}{=} \int_{\sqrt{2}/2}^0 t (1-t^2 + 1/5)^2 (-2t) dt = \frac{64}{175} \sqrt{2}, \\
\alpha_{2,1} &= -\frac{-\frac{128}{1575} \sqrt{2}}{\frac{64}{175} \sqrt{2}} = \frac{2}{9} \text{ Solution is: } [\alpha_{2,1} = \frac{2}{9}, \alpha_{2,0} = -\frac{11}{35}] \\
&\text{Deci } P_2(x) = x^2 + 2/9(x + 1/5) - 11/35 = x^2 + \frac{2}{9}x - \frac{17}{63} = 0, \text{ Solution is:} \\
&\frac{4}{63}\sqrt{7}\sqrt{10} - \frac{1}{9}, -\frac{4}{63}\sqrt{7}\sqrt{10} - \frac{1}{9} : -0.64232 : 0.4201
\end{aligned}$$

$$\begin{aligned}
A_0 &= \int_{-1}^1 \sqrt{1-x} \left(\frac{x - 0.4201}{-0.64232 - 0.4201} \right) dx = 1.1006 \\
A_1 &= \int_{-1}^1 \sqrt{1-x} \left(\frac{x + 0.64232}{+0.64232 + 0.4201} \right) dx = 0.78505
\end{aligned}$$

Caz particular $w(x) = 1 : \int_{-1}^1 f(x) = \sum_{i=0}^n A_i f(x_i) + R(f)$ e de tip Gauss dacă x_i sunt rădăcinile pol. lui Legendre de grad $n+1$:

$$P_{n+1}(x) = k_n \frac{d^{n+1}}{dx^{n+1}} \left((x^2 - 1)^{n+1} \right)$$

(Calcul Numeric, Bucur, Popeea, Simion).

Caz particular $w(x) = \frac{1}{\sqrt{1-x^2}}$, pe $[-1, 1]$ polinomul ortogonal de grad $n+1$ e Cebîșev : $T_{n+1}^*(x) = \frac{1}{2^n} \cos((n+1) \arccos x)$. nodurile $x_k = \cos \frac{(2k+1)\pi}{2(n+1)}$, $k = \overline{0, n}$.

$$A_k = \int_{-1}^1 \frac{T_{n+1}^*(x)}{(x - x_k) T_{n+1}'^*(x_k)} \frac{1}{\sqrt{1-x^2}} dx$$

3. Să se afle formula de cuadratură de grad de exactitate maxim de forma

$$\int_{-1}^1 f(x)dx = Af(-1) + Bf(x_0) + Cf(1) + R(f).$$

Figure 1:

se face sch. de var. $x = \cos t, t = \arccos x$

$$\begin{aligned} A_k &= \int_{\pi}^0 \frac{\cos((n+1)t)}{\left(\cos t - \cos \frac{(2k+1)\pi}{2(n+1)}\right) \left(-(n+1) \sin\left((n+1) \cos \frac{(2k+1)\pi}{2(n+1)}\right)\right)} (-dt) = \\ &= -\frac{1}{(n+1) \sin\left((n+1) \cos \frac{(2k+1)\pi}{2(n+1)}\right)} \int_0^{\pi} \frac{\cos((n+1)t) dt}{\cos t - \cos \frac{(2k+1)\pi}{2(n+1)}} = \end{aligned}$$

Am 4 const. de determinat, deci gradul maxim va fi ≥ 3 :

$$\begin{aligned} R(e_0) &= \int_{-1}^1 1dx - (A + B + C) = 0 \\ R(e_1) &= \int_{-1}^1 xdx - (A(-1) + Bx_0 + C) = 0 \\ R(e_2) &= \int_{-1}^1 x^2dx - (A + Bx_0^2 + C) = 0 \\ R(e_3) &= \int_{-1}^1 x^3dx - (A(-1)^3 + Bx_0^3 + C) = 0 \end{aligned}$$

Adică:

$$\begin{cases} A + B + C = 2 \\ -A + Bx_0 + C = 0 \\ A + Bx_0^2 + C = \frac{2}{3} \\ -A + Bx_0^3 + C = 0 \end{cases}$$

rezultă:

$$\begin{aligned} B(x_0^3 - x_0) &= 0 \\ x_0 &= 0 \\ A = C = \frac{1}{3}; B &= \frac{4}{3} \end{aligned}$$

f.c. va fi:

$$\begin{aligned}\int_{-1}^1 f(x) dx &= \frac{1}{3} (f(-1) + 4f(0) + f(1)) + R(f) \\ R(e_4) &= \frac{2}{5} - \frac{1}{3}(1+1) = -\frac{4}{15} \neq 0 \\ R(f) &= -\frac{4}{15} \frac{f^{iv}(\xi)}{4!}\end{aligned}$$

Ecuațiile parametrice ale unui segment care are capetele (x_1, y_1) și (x_2, y_2) :

$$\begin{aligned}x &= tx_2 + (1-t)x_1 \\ y &= ty_2 + (1-t)y_1\end{aligned}$$

unde $t \in [0, 1]$.

Cf. an 1:

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{y_2 - y_1}{x_2 - x_1}.$$

① Să se ceară ca dacă (P_n) este un șir de polinoame ortogonale atunci (P_n) verifică o relație de recurență de formă

$$P_{n+1}(x) = (a_n x + b_n)P_n(x) - c_n P_{n-1}(x), \quad n \geq 0, P_{-1} = 0$$

Dacă $P_n(x) = c_{0,n}x^n + \dots$ atunci

$$a_n = \frac{c_{0,n+1}}{c_{0,n}}, \quad c_n = \frac{a_n \int_a^b P_n(x) w(x) dx}{a_{n-1} \int_a^b P_{n-1}(x) w(x) dx}$$

Ideea: Polinoamele ortogonale formează o bază în mulțimea polinoamelor.

$$xP_n(x) = \sum_{k=0}^{n+1} c_k P_k(x) \quad (2)$$

înmulțim formula 1 cu $P_i(x)w(x)$ și integrăm de la a la b :

$$\begin{aligned}\int_a^b xw(x)P_n(x)P_i(x)dx &= \sum_{k=0}^{n+1} c_k \int_a^b P_k(x)P_i(x)w(x)dx \\ \int_a^b xw(x)P_n(x)P_i(x)dx &= c_i \int_a^b w(x)P_i^2(x)dx\end{aligned}$$

Dar $xP_i(x)$ este pol. de grad $i+1$, iar P_n este pol. de grad $n \perp$, deci integrala din dreapta este 0 pentru $i+1 < n$. Rezultă $c_i = 0$, pentru $i < n-1$. Formula

1 devine:

$$\begin{aligned} xP_n(x) &= \sum_{k=n-1}^{n+1} c_k P_k(x) = c_{n-1}P_{n-1}(x) + c_n P_n(x) + c_{n+1}P_{n+1}(x) \\ 0 &= c_{n-1}P_{n-1}(x) + (c_n - x)P_n(x) + c_{n+1}P_{n+1}(x) \end{aligned}$$

$$P_{n+1}(x) = (a_n x + b_n)P_n(x) - c_n P_{n-1}(x), \quad (3)$$

unde:

$$a_n := \frac{1}{c_{n+1}}, b_n := -\frac{c_n}{c_{n+1}}; c_n := -\frac{c_{n-1}}{c_{n+1}}$$

Din ultima formulă identificând coef, în (2), lui x^{n+1} :

$$\begin{aligned} c_{0,n+1} &= a_n c_{0,n} \\ a_n &= \frac{c_{0,n+1}}{c_{0,n}} \end{aligned}$$

Pentru det. lui c_n : înmulțesc formula 2 cu $P_{n-1}(x)w(x)$ și $\int_a^b$:

$$\begin{aligned} 0 &= a_n \int_a^b xP_n(x)P_{n-1}(x)w(x)dx - c_n \int_a^b P_{n-1}^2(x)w(x)dx \\ &\quad \int_a^b xP_n(x)P_{n-1}(x)w(x)dx = ? \end{aligned}$$

Dar $xP_{n-1}(x) = c_{0,n-1}x^n + q(x) = \frac{c_{0,n-1}}{c_{0,n}}P_n(x) + q_1(x)$, grad $q_1 \leq n-1$.

$$\begin{aligned} \int_a^b xP_n(x)P_{n-1}(x)w(x)dx &= \int_a^b P_n(x) \left(\frac{c_{0,n-1}}{c_{0,n}}P_n(x) + q_1(x) \right) w(x)dx = \\ &= \int_a^b \frac{c_{0,n-1}}{c_{0,n}} P_n^2(x)w(x)dx \end{aligned}$$

am obținut:

$$0 = a_n \frac{c_{0,n-1}}{c_{0,n}} \int_a^b P_n^2(x)w(x)dx - c_n \int_a^b P_{n-1}^2(x)w(x)dx.$$

③ Dacă $P_n(x_0) = 0$ nu se poate ca
 $\int_a^b \frac{P_n(t)}{t-x_0} w(t)dt = \frac{c_{0,n}}{c_{0,n-1} - P_{n-1}(x_0)w(x_0)}$

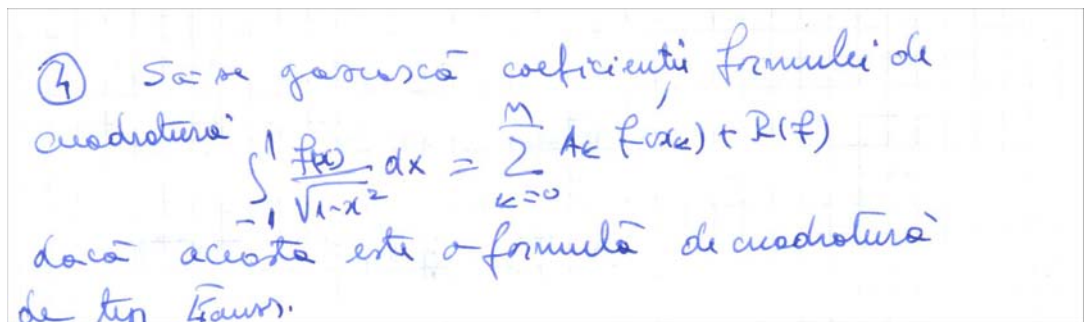
Ideea: Se calculeaza integrala folosind formula de tip Gauss cu nodurile rădăcinile polinomului P_n

$$\frac{P_n(t)}{t - x_k} = q(t)$$

q este polinom de grad $n - 1$, deci:

$$\int_a^b q(t) w(t) dt = \sum_{i=1}^n A_i q(t_i) = A_k q(x_k).$$

unde t_i sunt rădăcinile polinomului P_n .



Se știe că polinoamele ortogonale pe $[-1, 1]$ cu ponderea $\frac{1}{\sqrt{1-x^2}}$ sunt polinoamele lui Cebșev (au coef. lui $x^{n+1} = 1$):

$$T_{n+1}(x) = \frac{1}{2^n} \cos((n+1) \arccos x)$$

rădăcinile pol. lui Cebîșev:

$$x_k = \cos\left(\frac{2k+1}{2(n+1)}\pi\right), k = \overline{0, n}.$$

Formula de cuadratură va fi:

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = \sum_{k=0}^n A_k f(x_k) + R(f)$$

Iar

$$A_k = \int_{-1}^1 \frac{T_{n+1}(t)}{(t - x_k) T'_{n+1}(x_k) \sqrt{1-t^2}} dt.$$

În formula de la pr. 3 iau $P_n \rightarrow T_{n+1}$

$$\int_{-1}^1 \frac{T_{n+1}(t)}{(t - x_k) T'_{n+1}(x_k) \sqrt{1-t^2}} dt = \frac{1}{T'_{n+1}(x_k) T_n(x_k) \int_{-1}^1 T_n^2(x) \frac{1}{\sqrt{1-x^2}} dx}$$

$$\begin{aligned}
\int_{-1}^1 T_n^2(x) \frac{1}{\sqrt{1-x^2}} dx &= \int_{-1}^1 \frac{(\cos(n \arccos x))^2}{2^{2n-2} \sqrt{1-x^2}} dx \stackrel{\arccos x=t}{=} \\
&= \int_0^\pi \frac{\cos^2 nt}{2^{2n-2}} dt = \frac{1}{2^{2n-1}} \int_0^\pi (1 + \cos(2nt)) dt = \\
&= \frac{\pi}{2^{2n-1}}.
\end{aligned}$$

$$\begin{aligned}
T_n(x_k) &= \frac{1}{2^{n-1}} \cos \left(\left(n \arccos \left(\cos \frac{2k+1}{2(n+1)} \pi \right) \right) \right) = \\
&= \frac{1}{2^{n-1}} \cos \left(\frac{n(2k+1)}{2(n+1)} \pi \right) = \\
&= \frac{1}{2^{n-1}} \cos \left(\frac{2(n+1)k + (n+1) + (-2k)}{2(n+1)} \pi \right) = \\
&= \frac{1}{2^{n-1}} \cos \left(2k+1 - \frac{k}{n+1} \right) \pi
\end{aligned}$$

$$T'_{n+1}(x) = (n+1) \frac{-\sin((n+1) \arccos x)}{\sqrt{1-x^2}}$$

$$T'_{n+1}(x_k) = -(n+1) \frac{\sin \left((n+1) \frac{2k+1}{2(n+1)} \pi \right)}{\sqrt{1 - \cos^2 \frac{2k+1}{2(n+1)} \pi}}$$

Făcând calculele:

$$A_k = \frac{\pi}{n+1}.$$

c) Se notează cu polinoamele rădăcinile de grad n în raport cu produsul $w(x) = \sqrt{1-x^2}$, $w: [-1,1] \rightarrow \mathbb{R}_+$ sunt $U_n(x) = \frac{\cos(n \arccos(x))}{\sqrt{1-x^2}}$

U_n sunt polinoame de grad n , cf. Moivre:

$$(\cos t + j \sin t)^{n+1} = \cos(n+1)t + j \sin(n+1)t$$

$$\begin{aligned}
\sin(n+1)t &= \binom{n+1}{1} \cos^n t \sin t - \binom{n+1}{3} \cos^{n-2} t \sin^3 t + \dots \stackrel{t=\arccos x}{=} \\
&= \binom{n+1}{1} x^n (\sqrt{1-x^2}) - x^{n-2} (\sqrt{1-x^2})^3 + \dots
\end{aligned}$$

Ortogonalitatea:

$$\int_{-1}^1 \frac{\sin((n+1) \arccos x) \sin((k+1) \arccos x) \sqrt{1-x^2}}{1-x^2} dx = 0, k \neq n.$$

$$\stackrel{t=\arccos x}{=} - \int_0^\pi \sin((n+1)t) \sin((k+1)t) dt = 0, k \neq n.$$

⑦ Se se găsește formula de interpolare de grad maxim de exactitate de formă:

$$\int_{-1}^1 \sqrt{1-x^2} f(x) dx = \sum_{k=0}^n A_k f(x_k) + R(f)$$

Cf. 6 nodurile sunt rădăcinile polinomului U_{n+1} :

$$\begin{aligned} \sin((n+2) \arccos x) &= 0 \\ (n+2) \arccos x &= k\pi \\ \arccos x &= \frac{k\pi}{n+2}, k = \overline{1, n+1}. \\ x_k &= \cos \frac{k\pi}{n+2}, k = \overline{1, n+1} \end{aligned}$$

$$A_k = \int_{-1}^1 \sqrt{1-x^2} \frac{\sin((n+2) \arccos x)}{\sqrt{1-x^2} (x-x_k) U'_{n+1}(x_k)} dx$$